

Exploring circular water options for a water-stressed city: Water metabolism analysis for Paju City, South Korea

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ABSTRACT

For urban areas with limited water supplies, circular water strategies must be pursued to achieve water security goals. By tracking and balancing all water flows and their potential changes city-wide, urban water metabolism is a useful framework for exploring the effect of circular water strategies on performance. This study focused on a water-stressed city in South Korea, Paju. We first analyzed Paju's current and future water metabolism. We then explored the effects of 10 alternative water management options (implementing four circular water strategies separately or combined, in both conservative and maximum potential cases) on three aspects of water security (water resource efficiency, self-sufficiency, and supply diversification). In 2018, Paju had lower efficiency of water use than our benchmarking case, Seoul, despite Paju's higher level of supply self-sufficiency and diversity. By 2040, the expected growth and increasing water demand of the city would further increase its reliance on water imports, requiring expansion of water infrastructure. However, if Paju employs large-scale wastewater recycling and rainwater harvesting, and reduces its water loss and demand, the combined effect could offset the expected water supply increase between 2018 and 2040, as well as the associated need for additional infrastructure.

1. Introduction

Water security has become a popular concern for urban areas characterized by high population density, agglomeration of economic activities, and dependence on a hinterland for supplying resources. Previously, cities mainly focused on water supply issues, but the focus of urban water management to address water security has evolved and expanded. It includes the issues of sewerage and public health; flood protection and drainage; pollution management and environmental protection; water efficiency and productivity; and water-related risks, including adaptation to environmental changes, as well as social welfare and equity (Brown, Keath, & Wong, 2009; Hoekstra, Buurman, & Van Ginkel, 2018). Recently, the circular economy paradigm has been applied to the water sector, broadening the scope of water security strategies (IWA, 2016; Smol, Adam, & Preisner, 2020). In the concept of the circular water economy, water is not only a resource that provides products and services, but also acts as a carrier of materials and energy (Stuchtey, 2015; Tahir, Steichan, & Shouler, 2018). Therefore, the circular water economy emphasizes using water and water-related resources more efficiently and repeatedly in a closed loop, recovering

energy and nutrients, and preventing contamination of water in the first place. By retaining full values of water instead of discharging it after use, the circular use of water provides such benefits as reduced nutrient load to receiving waters; reduced energy consumption associated with production, treatment, and distribution of water; and improved agricultural production, if the treated water is used for irrigation (Voulvoulis, 2018).

Material flow analysis has been extensively adopted as a core method to explore and assess resource efficiency strategies for the circular economy. For example, to monitor the EU's progress toward a circular economy, Eurostat diagnoses each country's material flows and tracks relevant indicators, following the guidelines on economy-wide material flow accounts (Eurostat, 2001, 2018). An equivalent approach to direct water flows at an urban level is urban water metabolism (UWM) (Renouf, Sochacka, et al., 2017). Similar to material flow analysis, the strengths of the UWM include its holistic nature, which enables the tracking of all the anthropogenic and natural water flows across all stages of urban water management, from abstraction to discharges and losses (Renouf, Sochacka, et al., 2017). Traditionally, these water flows tend to be studied and managed separately at smaller scales for example, at a facility or basin. Therefore, performance indicators derived from the

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UWM are useful for strategic decisions regarding urban water goals (Renouf, Serrao-Neumann, Kenway, Morgan, & Low Choy, 2017). Water data required for this quantification can be easily obtained from publicly available statistics (Fragkou, Vicent, & Gabarrell, 2016). Integrated water modeling, such as Aquacycle, quantifies water flows based on water balance, but it simulates water flows usually at smaller scales, such as at the catchment or district level (Renouf & Kenway, 2017).

Compared to the sheer volume of material flow analyses on various resources, however, relatively little metabolism research has been conducted for water (Fragkou et al., 2016; Renouf, Serrao-Neumann, et al., 2017). Kenway, Gregory, and McMahon (2011) formalized UWM, emphasizing the importance of examining water reuse and losses in the water balance. By applying it to four Australian cities, the authors found that the cities used water inefficiently, leaving large flows of rainwater, stormwater, and wastewater untapped. Based on UWM, Fragkou et al. (2016) confirmed that 27 coastal municipalities in Catalonia receive enough rainfall for city-scale rainwater harvesting, and Paul, Kenway, McIntosh, and Mukheibir (2018) found significant potential for wastewater recycling and loss reduction in the water supply network in Bangalore. After reviewing various urban water goals expressed in the vision statement of water initiatives, Renouf, Serrao-Neumann, et al. (2017) proposed a set of UWM indicators across four objectives: resource efficiency (water, energy, and nutrients); supply internalization and diversification; protection of water resources and hydrological cycles; and recognition of the diverse functions of water.

UWM has also been adopted to screen the type and scale of alternative water management interventions needed to achieve water objectives. For a semi-hypothetical case, Farooqui, Renouf, and Kenway (2016) explored the impacts of six water servicing options on two objectives, namely, resource efficiency and hydrological performance. The study found that when harvesting rainwater and stormwater is adopted at large scale, it can reduce per capita extraction of external water, restore hydrological flows, and reduce water-related energy use. However, adopting these interventions at the household scale may incur an energy trade-off. Wastewater recycling can also bring similar water efficiency benefits, but has little impact on restoring hydrological flows. Renouf et al. (2018) examined the water efficiency and hydrological performance of four interventions for three city-regions of Australia: demand management, rainwater/stormwater harvesting, wastewater recycling, and increasing perviousness. The combined performance benefits were substantive, reducing between 10–36% of the per-capita water extraction, but the individual effect showed different priorities across three city-regions.

The UWM focuses mainly on describing the physical dimension of an urban water system. For a complementary understanding of the economic, environmental, and/or social aspects, UWM can be integrated with other evaluation approaches, such as cost accounting and life cycle assessment (Renouf & Kenway, 2017). Torres López, Barrionuevo, and Rodríguez-Labajos (2019) established a System of Environmental-Economic Accounting for water (SEEA-Water), for the first time at an urban level. The SEEA-Water not only records information on urban water flow in its physical supply and use table, which is equivalent to UWM, but also includes information on water quality (an emissions account) and monetary transactions (a hybrid account). Nezami, Zarghami, Tizghadam, and Abbasi (2022) introduced a hybrid model that combines UWM and system dynamics to evaluate water management options, in terms of the demand and supply gap as well as cost coverage. UWM can be a basis to conduct life cycle assessment for urban water systems, for example, to estimate their life cycle energy consumption (S. J. Kenway et al., 2015; Venkatesh, Chan, & Brattebø, 2014) or carbon emissions (Shehabi, Stokes, & Horvath, 2012; Xue et al., 2019). In particular, evaluating the carbon reduction potential of alternative water management options, such as reducing water losses or demand, initially requires quantitative information about relevant water flows from UWM.

In this study, we analyzed the current and future water metabolism

of Paju, a water-stressed city of Korea that has increased water imports and planned a large-scale infrastructure expansion to meet expected further increase in its water demand. Our objective is to examine the extent to which circular water options—wastewater recycling, rainwater harvesting, demand management, and loss reduction—could relieve Paju's reliance on water imports and the need for infrastructure expansion. To make plausible proposals for the city's water planning, we focused on improving the resolution of scenarios. For example, our scenarios considered detailed figures about the existing and planned capacities of individual water facilities, projections on the population, newly constructed buildings, and/or water demand increase, rather than introducing a single-parameter assumption, such as share of wastewater recycling. To evaluate Paju's water performance, we adopted a set of existing and new indicators tailored to local needs, with regard to water efficiency, self-sufficiency, and supply diversification. This is the first part of a larger study evaluating sustainability of water management at a whole-city scale, for the case of Paju. In subsequent studies, the results of this UWM will be used to evaluate the life-cycle energy use and carbon emissions of urban water management.

2. Data and method

2.1. Case study region

Our case study focused on Paju, which is among the cities in Korea that increasingly face water security issues. Furthermore, one of the authors had good access to data not publicly available. Located in the north-west of the greater metropolitan area and south of Panmunjeom on the 38th parallel, Paju is a growing city hosting Unjeong New Town and 14 industrial complexes (Fig. 1). Its land area is 673.21 km², which is 1.6 times that of Seoul, the capital and the largest metropolitan city of Korea, and its population surpassed 484,000 in early 2022 (MOIS, 2019; Paju City Government, 2022). However, Paju lacks local water resources. Its sole local source for surface water is the Imjin River, which originates from Masikryong in North Korea and flows to the Yellow Sea, crossing the demilitarized zone. As North Korea occupies 63% of its catchment area (MOE & K-water, 2021) and operates eight dams, Paju's water abstraction from the Imjin River has become less reliable (Cho & Jang, 2020). Instead, it has increased its water imports. As of 2018, Paju abstracted about 26.4 million m³ of water from Imjin River, but it imported an additional 92.2 million m³ of water from the metropolitan inter-regional water supply system (MOE, 2019c).

Despite the limited availability of local water sources, Paju's water demand is expected to increase further, requiring a large-scale expansion of water infrastructure. It is projected that, with the completion of the Unjeong New Town Housing Site Development Project, which began in early 2000 and will continue until 2023, Paju's population is likely to increase by 8% more in 2030 compared to that in 2018 (MOE, 2019c, 2021). In addition, the Paju Liquid Crystal Display industrial complex, one of the largest among the 14 industrial complexes in Paju, is planning a large-scale expansion in its production lines. This will increase both the industrial and residential water demand, owing to additional employment (KICOX, 2021; MOE, 2021; Paju City Government, 2018). Considering these factors, the city of Paju and the Ministry of Environment estimated that Paju will be short of 86,661m³ of water daily in 2040 (MOE, 2021). The effect of climate change is likely to further aggravate water security, making Paju one of the four most vulnerable municipalities in Gyeonggi Province by 2040 (Gyeonggi Province Government, 2017). In response to the expected increase in water stress, the city planned large-scale construction of an additional inter-regional water transfer grid as well as an industrial water production facility. Although inter-regional water supply systems have been widely adopted in Korea, this option must be carefully examined, because Paju has reached a limit in terms of further expanding its underground pipeline networks.

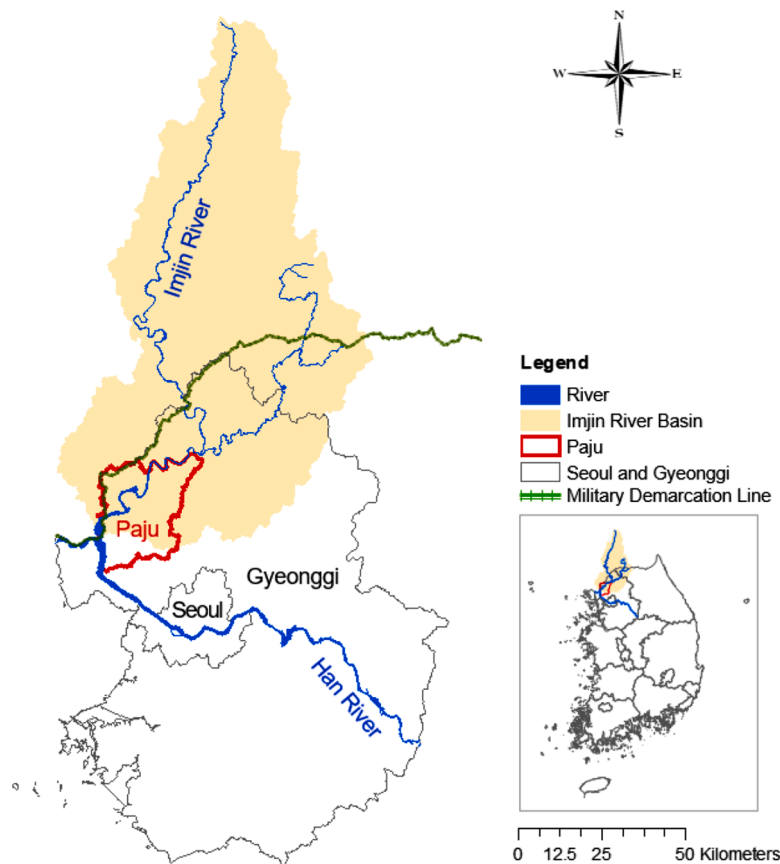


Fig. 1. Location of Paju in the greater metropolitan area and Imjin River basin

2.2. Urban water metabolism

We analyzed Paju's current and future water management using the UWM framework originally proposed by Kenway et al. (2011) and further developed by Renouf et al. (2018). The framework enables the holistic tracking of both anthropogenic and natural water flows across and inside an urban system boundary, by balancing inflows and outflows during all processes of the urban water system (Eq. 1). Anthropogenic water flows, which are managed by urban water infrastructure, include centralized water supplies (C) from surface water and groundwater, decentralized water supplies (D) harvested from precipitation or bore water, wastewater discharged (WW), and recycled flows (Re). Natural water flows include unharvested precipitation (P), stormwater runoff (SW), infiltration to groundwater (G), and evapotranspiration (ET) along the hydrological cycle.

$$P + C + D + Re = ET + SW + WW + G + Re + \Delta S \quad (1)$$

where:

P = unharvested precipitation falling within the system boundary (i.e., total precipitation excluding rainwater harvested within the system boundary)

C = total centralized water supply from surface water (C_s) and groundwater (C_g)

D = total decentralized water supply harvested from precipitation (D_p) and bore water (D_g)

ET = evapotranspiration from system area

SW = surface runoff discharged from the system area

WW = wastewater discharged from system area, excluding recycled flows

G = infiltration to groundwater

Re = reuse and recycling of wastewater

ΔS = change in stored water within the system

Our system boundary was set as the administrative boundary of Paju, because water resources are managed at a local and regional level, where the performance goals are established and data are collected (Jeong & Park, 2020). Paju's water flows were quantified for the reference years of 2018 and 2040 by compiling data from public statistics, field interviews, and literature (Table 1 and Section S1.1 in the Supplementary Material).

To evaluate Paju's water performance, we adopted seven indicators related to three aspects of water security: water resource efficiency, water self-sufficiency, and supply diversification (Table 2 and Section S1.2 in the Supplementary Material). Water resource efficiency relates to the productive use of water, whereas water self-sufficiency and supply diversification relate to the localization of water cycles by using various sources of water within cities (Rygaard, Binning, & Albrechtsen, 2011). These indicators were selected based on their relevance to the context and water issues of Paju after reviewing various water security indicators (ADB, 2020; Jensen & Wu, 2018; Van Leeuwen, Frijns, Van Wezel, & Van de Ven, 2012). For the purpose of our study, we introduced and adjusted three new indicators to our UWM analysis. Local water abstraction ratio evaluates the level of surface water or groundwater abstraction compared to the available water resources. Sourcing distribution represents a set of fractions of water according to various categories of sources. The diversity index or the Shannon Weaver index, which was first applied to quantify the diversity of the water systems at the state level in the US (Barakoti, Celik, Moorhead, & Apul, 2019), measures water source diversity in this study, with more detailed categories of water supply sources.

2.3. Water security effects of alternative water management strategies

To explore alternative water management options for enhancing

Table 1
Water flow data and their sources

Type	Flows	Sources	Year
Cs (Centralized water supply from surface water)	River water (residential, industrial), Tap water (residential), Precipitated water (industrial), Water imported, Water loss	Ministry of Environment Statistics of Waterworks (MOE, 2019c); 2040 Masterplan for the Supply of Inter-regional and Industrial Water (MOE, 2021); Estimated from Ministry of Environment Statistics of Waterworks (MOE, 2019c)	2018, 2040
	River water (industrial), Reservoir water (agricultural)	Water Resources Management Information System (Han River Flood Control Office, 2021)	2018
Cg (Groundwater)	Groundwater	Ministry of Environment Groundwater report (MOE, 2019a)	2018
Dp (Decentralized water supply harvested from precipitation)	Rainwater harvesting (residential)	Ministry of Environment Statistics of Sewerage (MOE, 2019b)	2018
	Rainwater (agricultural)	Water Resources Management Information System (Han River Flood Control Office, 2021)	2018
WW (Wastewater discharged from system area, excluding recycled flows)	Residential wastewater/sewage Treated wastewater, Infiltration/Inflow (I/I)	Ministry of Environment Statistics of Sewerage (MOE, 2019b); Estimated from Paju's Framework Plan for Sewerage Maintenance (Paju City Government, 2015a) and mass balance	2018, 2040
	Industrial wastewater, Treated wastewater	Ministry of Environment, Statistics on Operation of Public Wastewater Treatment Plants (MOE, 2019d); National Survey on Sources of Pollution (NIER, 2019); Internal data obtained from city of Gyeonggi Province; Estimated from the National Survey on Sources of Pollution (NIER, 2019) and mass balance	2018
	Greywater (residential), Wastewater reuse (for industrial and other uses, river maintenance)	Ministry of Environment Statistics of Sewerage (MOE, 2019b)	2018
	Industrial wastewater reuse	Ministry of Environment Statistics on	2018

Table 1 (continued)

Type	Flows	Sources	Year
P (Unharvested precipitation falling within the system boundary)	Greywater (industrial)	Operation of Public Wastewater Treatment Plants (MOE, 2019d)	2018
	Precipitation, Land area	Internal data obtained from interviews Korea Meteorological Administration Open Data Portal (KMA, 2021); Paju Statistical Yearbook (Paju City Government, 2020) Kim (2019)	2018
ET (Evapotranspiration from system area), G (Infiltration to groundwater), SW (Surface runoff discharged from the system area)	Evaporation, Groundwater infiltration, Surface runoff		2018
Balancing flows	Discharges (residential, treated water), Infiltrated water, Infiltration	Calculated from mass balance	2018, 2040

Table 2

Water security indicators used in the study (the equations' response to each indicator were taken from previous studies, modified based on them, or developed in this study: (a), (b), and (e) Jeong and Park (2020); (d) Renouf, Ser-
 rao-Neumann, et al. (2017); (g) Barakoti et al. (2019))

Water security dimensions	Indicators	Equations
Water resource efficiency	Water use intensity (m ³ /capita/year) Water loss rate (%)	$\frac{C + D + Re - \text{water loss}}{\text{population}}$ $\frac{\text{water loss}}{(C + D + Re)}$
Water self-sufficiency	Local water abstraction ratio (%) Supply internalization (%) Water import rate (%)	$\frac{C}{\text{water available}}$ $\frac{(D + Re)}{(C + D + Re)}$ $\frac{\text{water import}}{(C + D + Re)}$
Supply diversification	Sourcing distributions (%) Diversity index (Shannon Weaver index)	$\frac{\text{water supply from each source}}{(C + D + Re)}$ $-\sum_{i=1}^N p_i \ln p_i \text{ where } p_i \text{ is the fraction of water allocated among various categories of sources}$

Paju's water security by 2040, we constructed 10 scenarios according to the type of circular water strategies (i.e., wastewater recycling, rain-water harvesting, reduction of water loss, water demand management, and all combined) and their size of effect (i.e., conservative and maximum potential) (Table 3). Among diverse circular water strategies (IWA, 2016), the study particularly focused on four water management strategies that are commonly practiced in the context of Korea. The conservative scenarios were constructed based on water goals that Paju has already established, or considering the minimum potential of alternative water management, whereas maximum potential scenarios were constructed based on the maximum potential or higher efficiency of alternative strategies. Detailed descriptions and the rationale for the 10 scenarios are presented in the Supplementary Material (Section S1.3 in the Supplementary Material).

Table 3
Ten scenarios for alternative water management in Paju in 2040

Water management options	Conservative Scenarios	Maximum Potential Scenarios
(1) Wastewater recycling	(S1-1) Use of treated wastewater as industrial water to the 2030 level set by the city's wastewater recycling plan (Paju City Government, 2015a). This would reduce industrial water import.	(S1-2) In addition to S1-1, further expand the use of treated wastewater to meet an expected increase in industrial water demand of Paju LCD industrial complex, which is estimated by the "2040 Masterplan for the Supply of Inter-regional and Industrial Water" (MOE, 2021).
(2) Rainwater harvesting	(S2-1) Use of rainwater for non-potable residential applications (e.g., washing, landscaping, toilet) to the 2020 level set by the city's water reuse plan (Paju City Government, 2015b). This would reduce upstream residential water import and groundwater use.	(S2-2) Use of rainwater for non-potable residential applications (e.g., washing, landscaping, toilet) to the level equivalent to 100% of Paju's rainwater harvesting potential, which was estimated based on existing and potential rainwater harvesting facilities, monthly rainfall events, and average harvesting ratio (Paju City Government, 2015b).
(3) Reduce water loss	(S3-1) Reduce water loss to 4.0%, which in turn would reduce residential water import	(S3-2) Reduce water loss to 2.5%, the benchmarking rate observed in Seoul (the loss rate in 2015 estimated by Jeong and Park (2020))
(4) Reduce water demand	(S4-1) Adopt water-efficient appliances that reduce water use by 20% (Kim & Oh, 2017; Yun & Cho, 2018) in newly constructed residential buildings between 2025 and 2040 (MOE, 2021; MOIS, 2022). This would reduce the upstream residential water import and groundwater use.	(S4-2) Utilize water-efficient appliances that reduce water use by 20% in all residential buildings in Paju
(5) All scenarios combined	(S5-1) Adopt all conservative scenarios 1-4	(S5-2) Adopt all maximum potential scenarios 1-4

3. Results and discussion

3.1. Paju's water metabolism for 2018 and 2040

The water metabolism of Paju (Fig. 2) showed that the city consumed 360 million m³ of water in 2018. Of the total, 39.2% was used in industry, followed by the agriculture (34.5%) and the residential sector (23.2%). On a per capita basis, this level of annual water consumption was equivalent to 777 m³ (Table 4), which was 6.1 and 1.4 times higher compared to that of Seoul (the metropolitan city) and Ulsan (the largest industrial city of Korea), respectively, as reported by Jeong and Park (2020). A major factor that increased the level of water consumption in Paju was the industrial water demand from the electronic component production company. In 2018, the facility required 129 million m³ of water, including 70 million m³ of greywater and 9 million m³ of treated wastewater. The level of greywater and wastewater reuse in this facility alone, was higher than the annual residential water consumption in the entire city, 83 million m³. Paju's per capita residential water use was 64% and 24% higher than that of Seoul and Ulsan, respectively. Paju's water loss rate was higher than that of Seoul and had additional discharges from decentralized wastewater systems in the rural areas of the residential sector. In particular, the decentralized wastewater discharge, either from individual septic tanks or more advanced treatment systems, was significant in Paju, constituting 68% of the total residential wastewater. As a result, Paju's centralized water supply and groundwater abstraction were higher than those of Seoul and Ulsan. In particular, Paju's groundwater use comprised 11% of the entire Gyeonggi province,

which was the second highest among all the municipalities in the province (MOE, 2019a).

Our analysis of Paju's water metabolism also confirmed that in 2018, Paju showed higher reliance on water imports (26.7%) than Ulsan (21.3%), a water-stressed industrial city. Simultaneously, Paju increased the level of alternative water sourcing (38.6%) to address the challenges of water security. In 2018, Paju's supply internalization was 24.6%, which was 5 times higher than that of Ulsan, and diversity index was 1.82, higher than that of Ulsan (1.38). This was particularly driven by the reuse of treated wastewater and greywater for industries, which was the second largest in Korea (MOE, 2019b). In Paju, more than 23% of the treated wastewater was directed toward industrial use (compared to 1% at the national level) and greywater reuse in industries accounted for 16% of the total greywater use at the national level (MOE, 2019b). This implies that Paju was in a relatively advanced position in Korea, promoting water reuse and supply internalization.

According to Paju's water metabolism for 2040 (Fig. 3), its annual water demand will increase to 418 million m³, particularly driven by an increase in industrial water demand (by 24.7%) and residential demand (by 23.1%). Industries would constitute 42.8% of the total water demand, while agricultural and residential uses would comprise 29.9% and 24.8%, respectively. Accordingly, the per capita annual water demand would also increase from 777 m³ to 808 m³. At present, Paju's plan to meet this increasing water demand is by expanding water imports (6.8 million m³/year of industrial water and 22 million m³/year of residential water) from inter-regional water supply systems and to construct a water production plant that supplies industrial water. This plan would increase Paju's reliance not only on the local water abstraction ratio, but also on water imports from 26.7% in 2018 to 35.9% by 2040, decreasing Paju's water supply diversity index from 1.82 in 2018 to 1.74 in 2040. This implies that the water security of Paju is expected to degrade by 2040, making the city more vulnerable to regional water conflicts and the impact of climate change (Kim, 2021; Li, Zuo, & Zhang, 2022).

3.2. Water security effects of alternative water management strategies

Table 5 and Table S2-3 in the Supplementary Material present the effect of 10 alternative water management options on the total water supply, local water supply, and water imports, and the selected performance indicators. Among the four strategies, the expansion of wastewater recycling showed the largest potential to increase local water sourcing and reduce water imports in both conservative and maximum potential scenarios. According to S1-2, reuse of wastewater, along with a slight increase in local abstraction, would reduce approximately 36 million m³ of water imports per year, which is equivalent to 65.3% of the increase in water supply between 2018 and 2040 (i.e., 55 million m³/year), or 24.0% of water imports estimated for 2040. In turn, this would increase Paju's supply internalization from 33.5% in the base case to 41.8%, and the diversity index from 1.74 to 1.86, while decreasing the water import rate from 35.9 to 27.3%.

While wastewater recycling would not change the level of total water demand and supply, water demand management would reduce it to the highest level among the four strategies. In the case of S4-2, 21.46 million m³ of water supply would be reduced, most of which would occur in the water import reduction. Therefore, the water import rate would decrease from 35.9 to 33.6%. The proportion of groundwater used for residential purposes and local water abstraction ratio would decrease from 8.1 to 7.3%, and 53.2 to 49.3%, respectively, the lowest among all scenarios. Accordingly, Paju's per-capita residential water use would decrease from 200 m³ in the base case, but it would still be 1.5 times higher than that of Seoul.

On the other hand, the expansion of rainwater harvesting showed the lowest potential for increasing alternative water supplies and reducing water imports. Even in the case of the maximum potential scenario S2-2, only 729,000 m³/year could be harvested from rainwater, resulting in

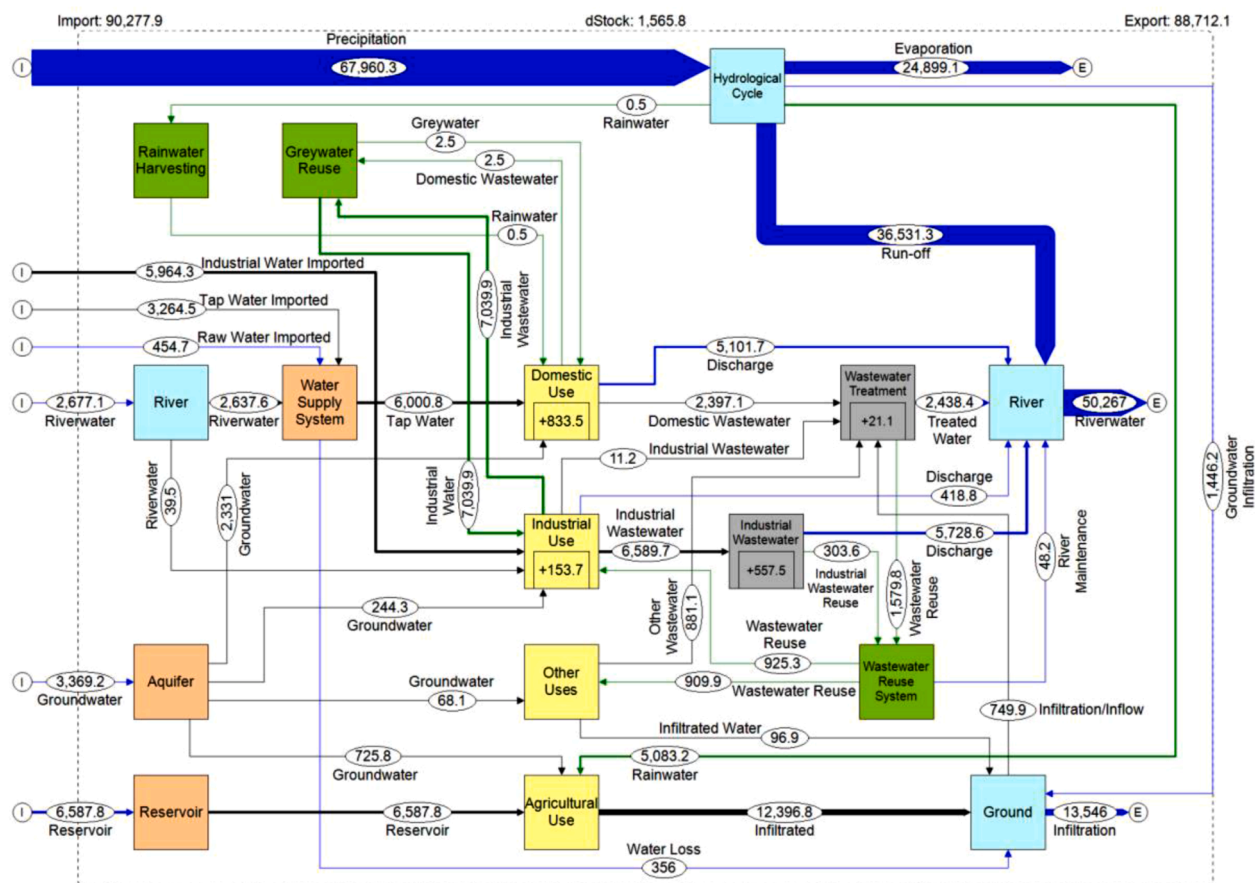


Fig. 2. Paju's water metabolism for the year 2018 (Unit: 10,000 m³/year)

Table 4

Water security indicators for Paju in 2018 and 2040 (* Values for Seoul and Ulsan were obtained directly from Jeong and Park (2020) or calculated based on their results.)

Water security indicators	Paju 2018	Seoul* 2040	Ulsan* 2015	2015
Water use intensity (m ³ /capita/year)	777	808	127	536
Residential	180	200	110	145
Industrial	307	346	4	267
Agricultural (Excluding effective rainfall)	268 (158)	242 (143)	1 (1)	114 (72)
Other	22	20	12	10
Water loss rate (%)	5.6	5.6	2.4	6.7
Local water abstraction ratio (%)	51.6	53.2	50.4	-
Supply internalization (%) (Excluding effective rainfall)	38.6 (24.6)	33.5 (21.3)	9.3 (9.2)	12.6 (5.0)
Water import rate (%)	26.7	35.9	5.9	21.3
Sourcing distributions	River 7.4% Reservoir 18.1% Groundwater 9.2% Rainwater 14.0% Greywater 19.4% Wastewater 5.2% Water imports 26.7%	River 6.9% Reservoir 15.7% Groundwater 8.1% Rainwater 12.1% Greywater 16.8% Wastewater 4.5% Water imports 35.9%	River 82.8% Reservoir 0.3% Groundwater 1.6% Rainwater 0.1% Greywater 0.3% Wastewater 8.9% Water imports 6.0%	Dam 57.3% Reservoir 12.0% Groundwater 7.8% Rainwater 7.6% Greywater 2.5% Wastewater 2.0% Water imports 10.8%
Diversity index	1.82	1.74	0.65	1.38

insignificant changes in the performance indicators. Reducing water loss, as represented in S3-1 and S3-2, also caused minimal changes in the water supply volumes and performance indicators. Although Paju's water loss rate was higher than that of Seoul, it was lower than the national average (10.8%)

When all the four strategies were combined, it was estimated that a

conservative scenario would increase alternative water sourcing by 6.87 million m³ (7.7% of total alternative sourcing in 2040), while reducing water imports by 9.03 million m³ (6.0% of water imports in 2040). On the other hand, our maximum potential scenario showed a considerably higher potential, increasing alternative water sourcing by 35.58 million m³ (39.9%) and reducing water imports by 55.68 million m³ (37.7%).

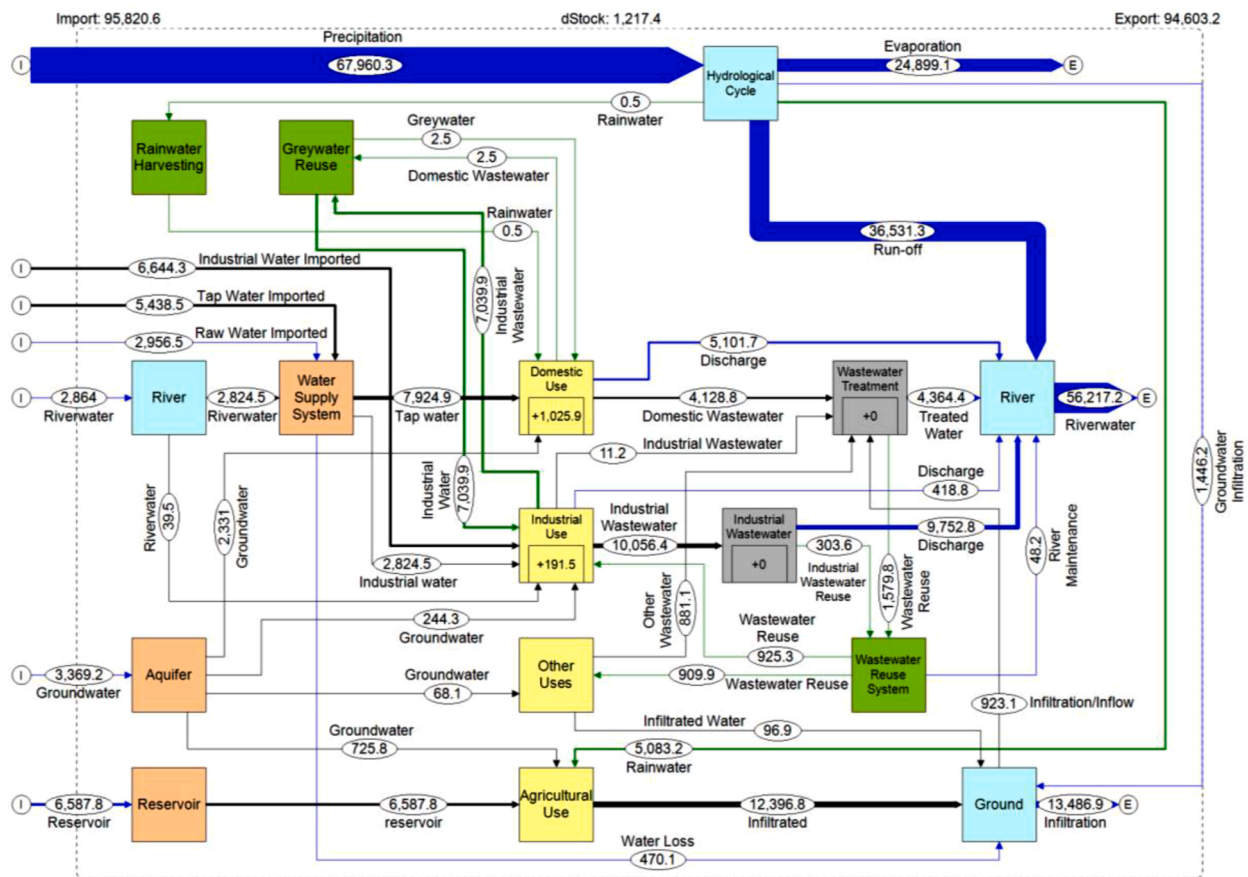


Fig. 3. Paju's water metabolism for the year 2040 (Unit: 10,000 m³/year)

Such reduction was equivalent to the water supply increase between 2018 and 2040, implying that the four strategies, in combination, would have the potential to offset the effect of the water demand increase during this period.

3.3. Water management proposals for Paju

The study focused on making plausible proposals, which are derived from our findings as follows. First, Paju needs to expand the use of treated wastewater to reduce risks related to water import and reliance on large-scale infrastructure expansion. Paju's water infrastructure is aging and has already reached a limit in expanding underground pipeline network. If approximately 34% of the extended wastewater is reused (as in S1-2), this alone would be able to offset 65.3% of the expected water supply increase between 2018 and 2040 and avoid the construction of an industrial water production facility. A benchmarking case can be found in Ulsan. To address challenges arising from increasing demand for industrial water while facing a supply constraint due to the deteriorating quality of imported water, Ulsan has promoted wastewater reclamation projects since 2019. As a result, 6.13 million m³/year of treated wastewater from a public industrial wastewater treatment is subjected to further treatment and supplied to seven industrial facilities nearby (BK-ENG, 2022). This is equivalent to the level envisioned in Paju's original plan (Paju City Government, 2015a) and the scenario S1-1. Considering that Paju generates an enormous volume of wastewater (141 million m³/year) and faces growing demand from water-intensive industries, wastewater recycling is a potential option that has a significant impact on its water security.

Second, Paju needs to prioritize its efforts with regard to reducing water demand. Paju's water use efficiency was still relatively lower, as indicated by its residential water use intensity, which was 1.6 times

greater than that of Seoul. If all households in Paju adopt water-saving appliances at 20% efficiency, this would reduce 30% of the extended water supply increase between 2018 and 2040. Considering that the maximum water saving rate of 50–60% has been reported according to the type of water-efficient appliances (Cho, Kang, & Eonku, 2011; Yun & Cho, 2018), the potential of water demand management options may be even larger. The current "Water Supply and Waterworks Installation Act" requires the installation of water-efficient appliances in newly constructed buildings, and recently adopted mandatory water rating labeling schemes for toilets and water faucets in some installations. To promote the effective adoption and use of water-efficient appliances, Paju needs to consider introducing subsidies for these appliances in existing buildings, and promoting a "Carbon Point System" that provides monetary incentives to those who reduce their consumption of electricity, tap water, and city gas (Gyeonggi Province Government, 2021).

Third, rainwater harvesting and reducing water loss would have only a limited impact; therefore, its feasibility and ways for improvement need to be carefully examined. One of the reasons for low rainwater harvesting potential was that the installed capacity of rainwater tanks required by the "Act on Promotion and Support of Water Reuse" was initially not significant (Kim, 2014). As a survey reported a limited water demand as one of the barriers in existing rainwater harvesting facilities in Seoul (Seoul Metropolitan Government, 2013), Paju needs to consider potential water demand, in addition to total floor area, when selecting target buildings and designing rainwater harvesting facilities. Strategically sizing a rainwater tank by building class and diversifying applications is critical to ensure economic feasibility of the rainwater harvesting facilities (Mun, Kim, Park, Lee, & Kim, 2012). Van Dijk, Lounsbury, Hoekstra, and Wang (2020) argued that such strategic and tailored designing of rainwater harvesting systems is needed, rather than uniformly applying one-size configuration to all buildings. This would

Table 5
The results of 10 scenarios

Water flow changes (in 10,000 m ³ /year)	Selected performance indicators		Water import rate	Sourcing distributions (%)	Diversity index
	Local water abstraction ratio	Supply internalization			
(S1-1)	53.2	35.0	34.3	River 6.8 Reservoir 15.7 Groundwater 8.0 Rainwater 12.1 Greywater 16.8 Wastewater 6.1 Water imports 34.3	1.77
(S1-2)	54.4	41.8	27.3	River 7.2 Reservoir 15.7 Groundwater 8.0 Rainwater 12.1 Greywater 16.8 Wastewater 12.8 Water imports 27.3	1.86

(continued on next page)

Table 5 (continued)

Water flow changes (in 10,000 m ³ /year)	Selected performance indicators				
	Local water abstraction ratio	Supply internalization	Water import rate	Sourcing distributions (%)	Diversity index
(S2-1)	53.2	33.5	35.9	River 6.8 Reservoir 15.7 Groundwater 8.0 Rainwater 12.2 Greywater 16.8 Wastewater 4.5 Water imports 35.9	1.74
(S2-2)	53.1	33.6	35.8	River 6.8 Reservoir 15.7 Groundwater 8.0 Rainwater 12.3 Greywater 16.8 Wastewater 4.5 Water imports 35.8	1.74

(continued on next page)

Table 5 (continued)

Water flow changes (in 10,000 m ³ /year)	Selected performance indicators			Sourcing distributions (%)	Diversity index
	Local water abstraction ratio	Supply internalization	Water import rate		
(S3-1)	53.2	33.6	35.7	River 6.9 Reservoir 15.8 Groundwater 8.1 Rainwater 12.2 Greywater 16.9 Wastewater 4.5 Water imports 35.7	1.74
(S3-2)	53.2	33.7	35.5	River 6.9 Reservoir 15.8 Groundwater 8.1 Rainwater 12.2 Greywater 16.9 Wastewater 4.5 Water imports 35.5	1.74

(continued on next page)

Table 5 (continued)

Water flow changes (in 10,000 m ³ /year)	Selected performance indicators		Water import rate	Sourcing distributions (%)	Diversity index
	Local water abstraction ratio	Supply internalization			
(S4-1)	53.0	33.5	35.8	River 6.9 Reservoir 15.8 Groundwater 8.0 Rainwater 12.2 Greywater 16.9 Wastewater 4.5 Water imports 35.8	1.74
(S4-2)	49.3	35.3	33.6	River 7.2 Reservoir 16.6 Groundwater 7.3 Rainwater 12.8 Greywater 17.7 Wastewater 4.7 Water imports 33.6	1.76

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Table 5 (continued)

Water flow changes (in 10,000 m ³ /year)	Selected performance indicators Local water abstraction ratio	Supply internalization	Water import rate	Sourcing distributions (%)	Diversity index
(S5-1)	53.0	35.3	34.0	River 6.9 Reservoir 15.8 Groundwater 8.0 Rainwater 12.3 Greywater 16.9 Wastewater 6.1 Water imports 34.0	1.77
	50.2	44.5	24.0	River 7.6 Reservoir 16.7 Groundwater 7.3 Rainwater 13.1 Greywater 17.8 Wastewater 13.6 Water imports 24.0	1.87

maximize two critical advantages of decentralized infrastructure—multi-functionality and high adaptability. Also, to overcome reported barriers such as lack of professional operators and high maintenance cost (Seoul Metropolitan Government, 2013), a larger-scale adoption (Farreny, Gabarrell, & Rieradevall, 2011; Lee, Kim, Kang, & Jeong, 2012) and provision of necessary maintenance services may be required.

The reduction of water loss also had a low impact because of Paju's efforts to reduce water loss since 2011 (Choi, 2019). Paju achieved a decrease in the water loss rate from 10.2% in 2011 to 5.6% in 2018 (MOE, 2012, 2019c). A portion of the efforts focused on replacing and managing water pipeline, which requires about 20 million USD investment to reduce 1% of water loss rate. In addition, Paju could further reduce water loss through sensor-based monitoring, which was adopted as a part of the "Smart Water City" pilot project initiated in 2014 (Lee & Lee, 2020). However, considering that more than 20% of Paju's pipelines are older than 20 years (MOIS, 2021), Paju would need to manage its infrastructure regularly to maintain a low water loss rate or reduce it further.

3.4. Limitations of the study

Using UWM, our study evaluated Paju's water management holistically at an urban level. While our UWM study incorporated most of the available data to the best extent possible, some flow estimates were more uncertain due to the varying data quality and assumptions introduced. When evaluating the uncertainty of Paju's water flows for the year 2018 using a pedigree-based data quality assessment method (Laner, Feketitsch, Rechberger, & Fellner, 2016; Weidema & Wesnæs, 1996), two water flows—decentralized residential wastewater discharges and infiltration/inflow to the public wastewater treatment facilities—particularly showed higher level of uncertainty (Table S2-1 in the Supplementary Material). Regarding the residential wastewater that is directly discharged from two types of decentralized wastewater systems in Paju, individual septic tanks or more advanced decentralized treatment systems, there were no monitoring data available with regard to the actual quantity of treatment or the quality of the discharged water. While UWM allowed us to estimate this figure from mass balance, which turned out to be within the range of installed capacity values reported, the estimate needs to be reviewed and verified. Surveys need to be conducted to monitor the effectiveness of decentralized system operation and any cases of failures. Estimates used for Paju's future water flows and scenario analyses inherently introduced additional uncertainties. While our study utilized reliable estimates from government reports and/or planning documents to the best extent possible, future studies can adopt different methods to explore future water cycles and scenarios.

In exploring alternative water management, the study mainly focused on the metabolic effect of four circular water strategies on water security by considering direct water flows only. To further understand the feasibility and sustainability of the strategies, future studies need to analyze life-cycle cost, associated nutrient flows and water quality, and/or carbon emissions. For example, Yerri and Piratla (2019) evaluated the life-cycle costs and expected benefits of onsite and satellite decentralized greywater reuse systems, and Woltersdorf, Scheidegger, Liehr, and Döll (2016) quantified nutrient and salt flows associated with water reuse for agriculture irrigation, using mathematical material flow analysis. Also, several studies evaluated the carbon emissions of decentralized water reuse systems, greywater use, and rainwater harvesting, considering reducing stormwater runoff, avoiding upstream/downstream treatments and/or reducing contaminant discharges (Chang, Lee, & Yoon, 2017; Kavvada et al., 2016). Similar analyses need to be conducted to assess the extent to which circular water strategies increase or decrease the life-cycle cost, nutrient discharges, and/or carbon emissions to better understand their sustainability in a specific local context.

4. Conclusions

This study examined the effect of four circular water strategies on three water security goals for the city of Paju, South Korea. We adopted the UWM framework and derived indicators of water efficiency, water self-sufficiency, and supply diversification. This study has two contributions. First, to present plausible proposals implementable for the case study region, we compiled and reflected, in the form of scenarios, the detailed information regarding the capacity of the existing and planned construction of water infrastructure, as well as projections on population, newly constructed buildings, and/or water demand increase. Second, in addition to the water metabolism indicators used in previous studies, we adopted three indicators (the local water abstraction ratio, sourcing distribution, and diversity index) to better represent the local contexts for water security.

The analysis of Paju's water metabolism for the year 2018 showed that the city promoted the use of diverse water sources in response to local water constraints, but still had lower water efficiency than a benchmarking metropolitan city, Seoul. To meet the increasing water demand by 2040, the city is planning to rely primarily on the expansion of the water supply infrastructure, which would lead to an increase in the local water abstraction ratio and water import rate, while decreasing supply internalization and diversity. Instead, if Paju implements four circular water options, namely, wastewater recycling, rainwater harvesting, water loss reduction, and water demand management, the city could avoid water imports, equivalent to the expected increase in water supply between 2018 and 2040 in the best case. Water recycling showed the highest potential, followed by water demand management, whereas rainwater harvesting showed a limited impact. This study explored circular options to manage water as a resource from a metabolic perspective, but future studies are needed to evaluate other sustainability aspects, such as water quality, economic costs, environmental consequences, and/or water-related risks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2022.104355.

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